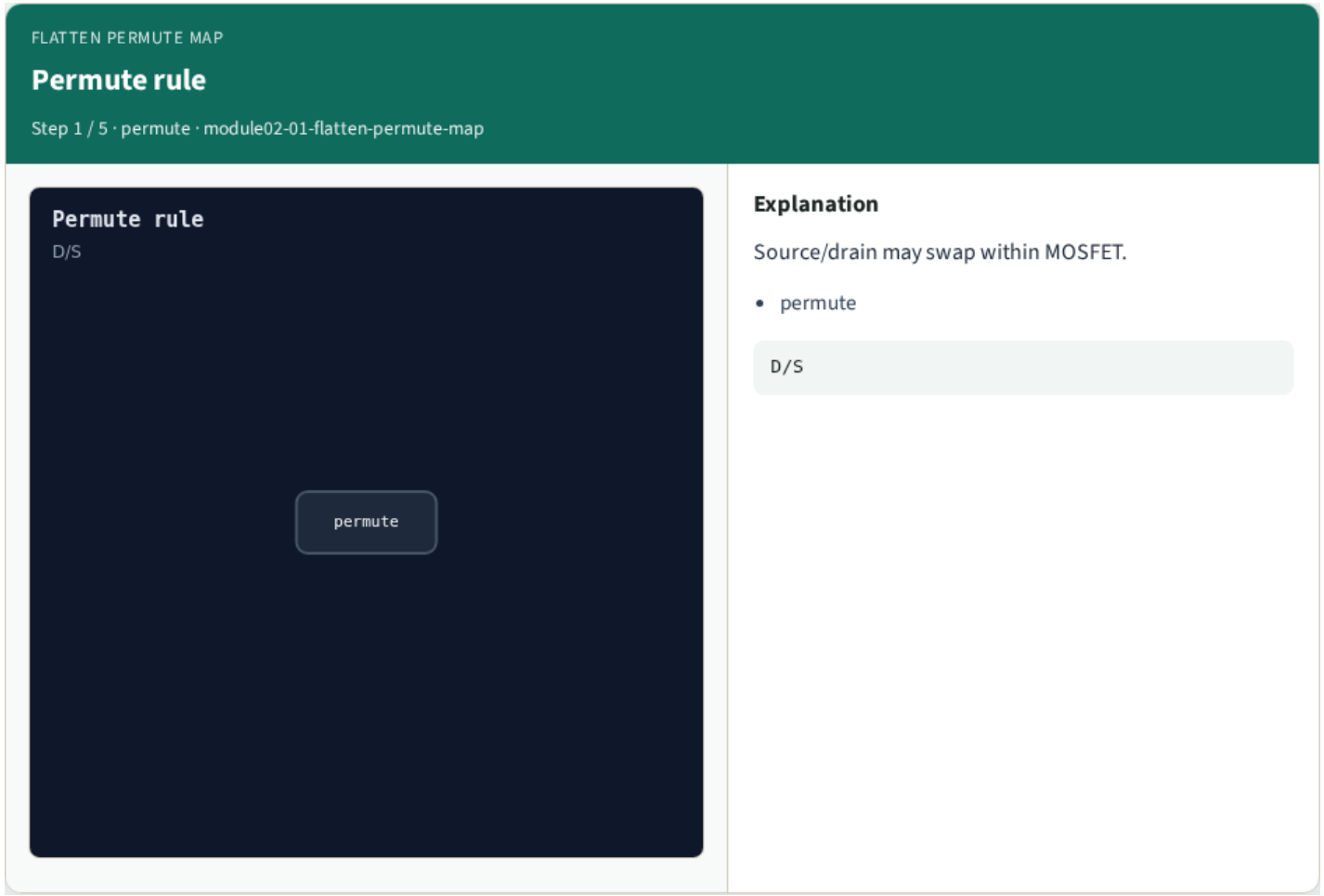
Why Flatten

Source/drain pins may permute within a MOSFET class

The idea

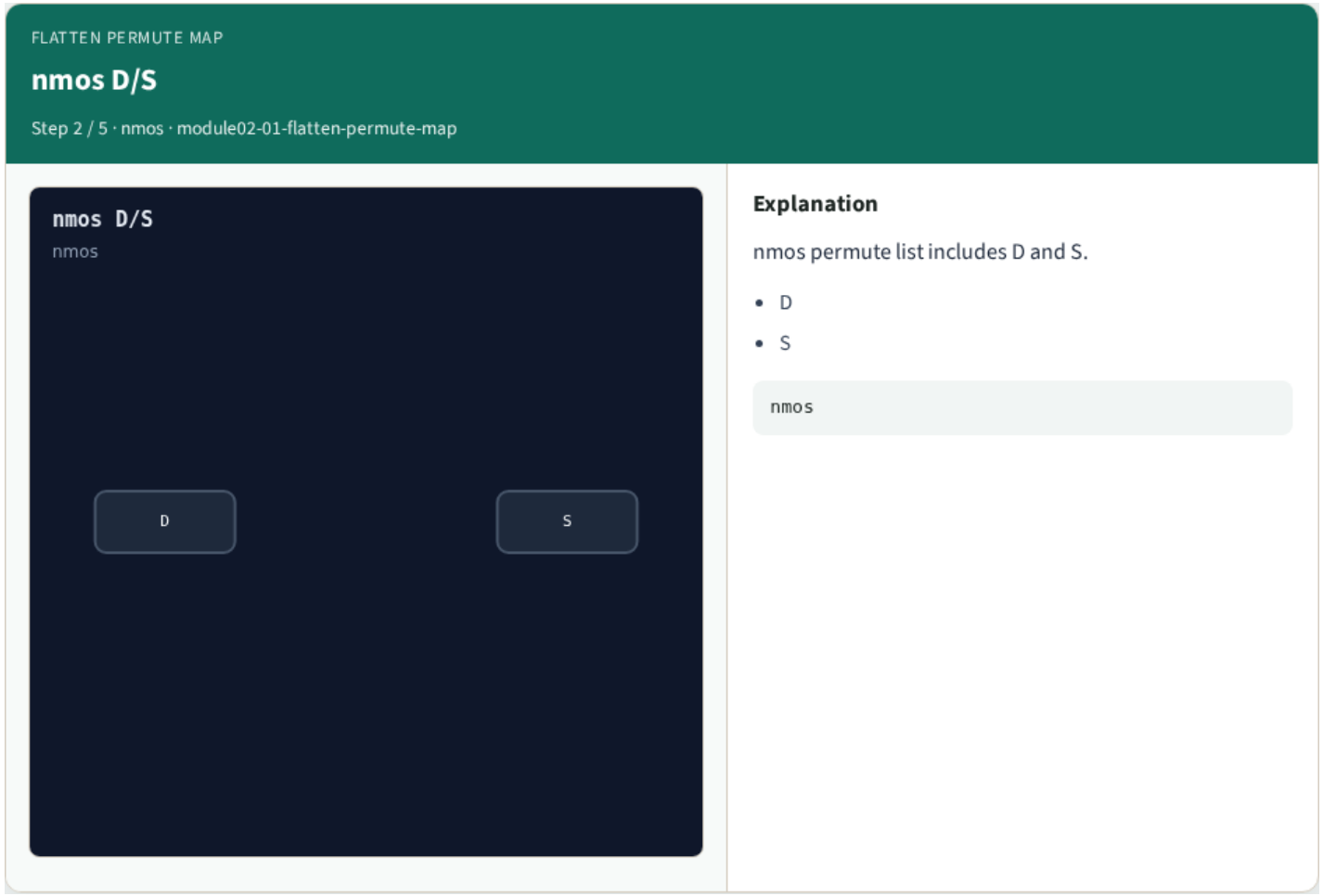
- GOLDEN: permute D/S on nmos/pmos

Permute rule



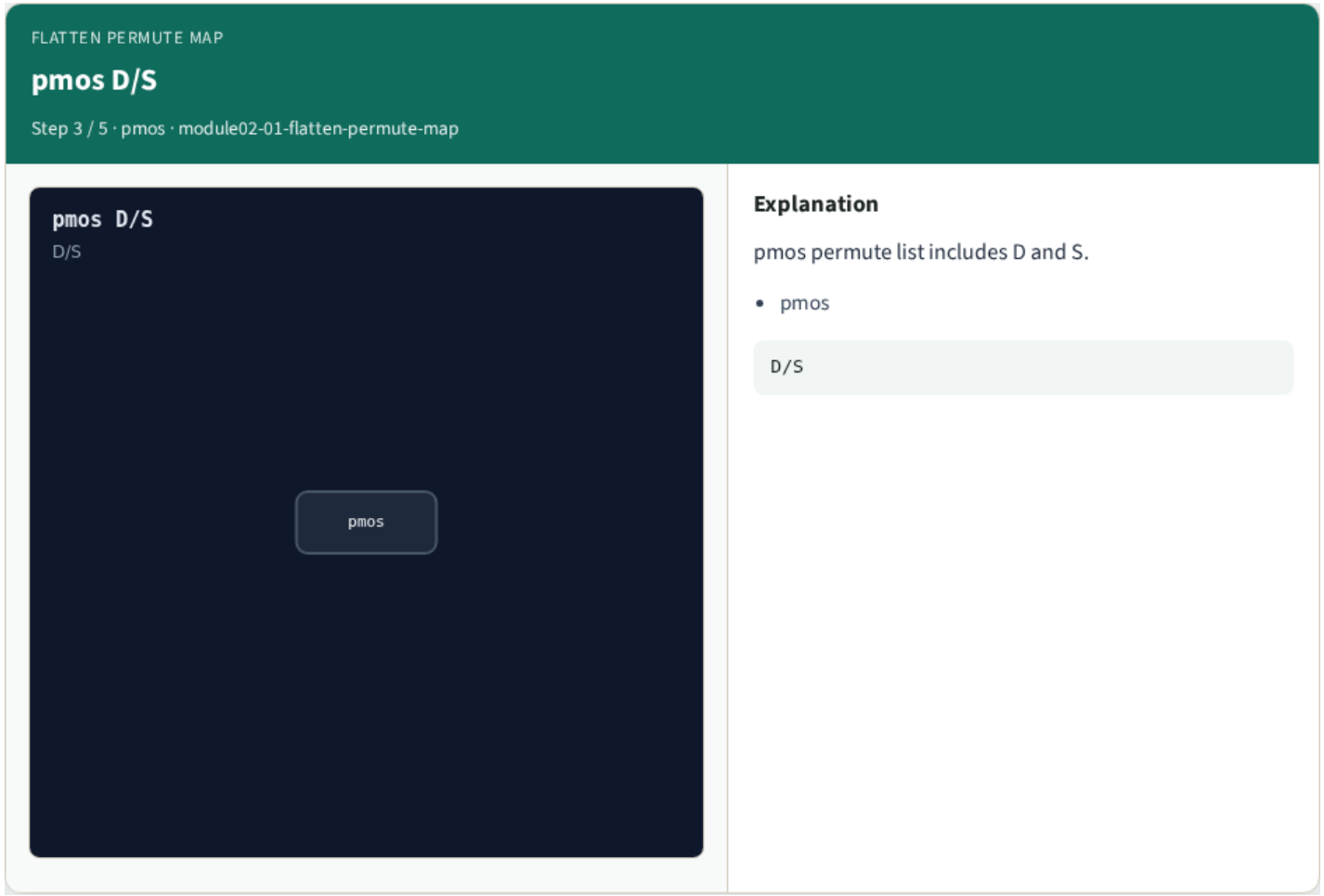
Permute rule

nmos D/S



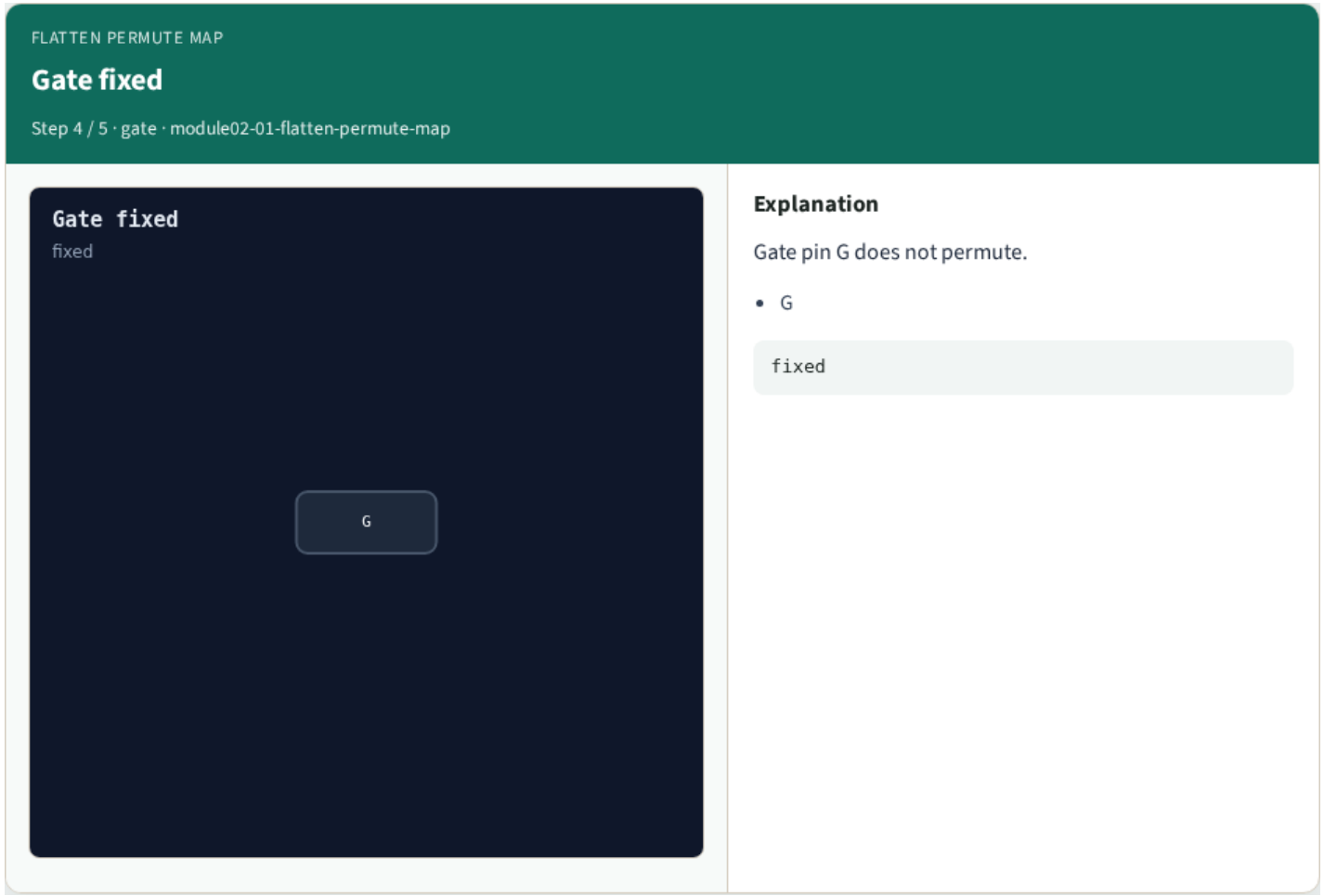
nmos D/S

pmos D/S



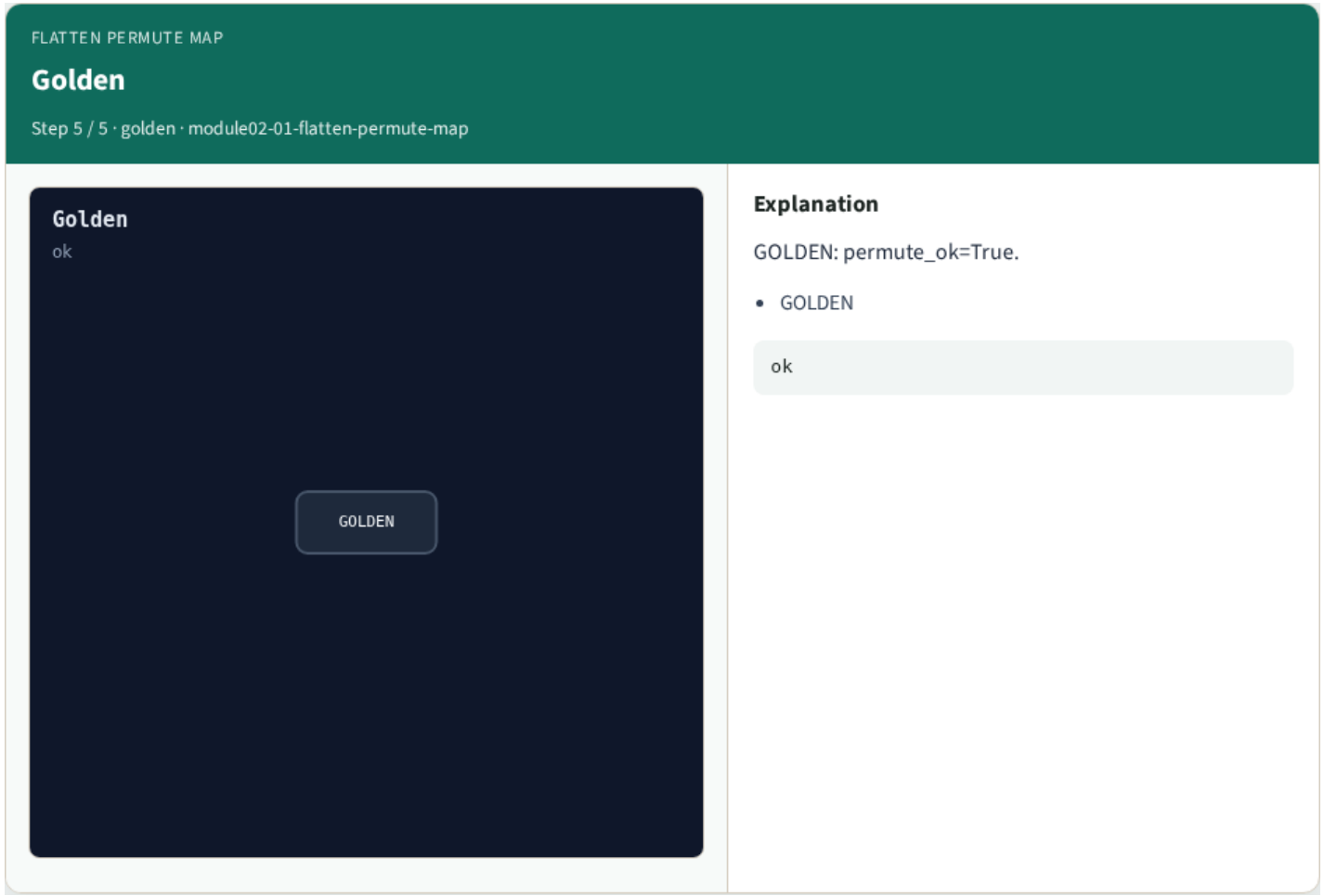
pmos D/S

Gate fixed



Gate fixed

Golden



Golden

Browser lab track

- Open **flatten-permute-map**
- Run the stage buttons and Check the goldens

Implement track

- From common/, run solvers.py and confirm NETGEN_PARTA_OK

Pitfalls

- Confusing schematic SPICE with layout extract
- Forgetting permute on source/drain

Try these

- From the common folder, run the Netgen map self-test

Try these — try these

```
cd courses/learn_netgen/common
python3 solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
permute D/S on nmos/pmos.
ok if goldens match
```

Next

- Complete the quiz for this part
- Continue the Netgen Part A map, then the real CLI track